

ARYAN GUPTA

AI Engineer — GenAI Systems — LLM Applications — FastAPI · RAG · AWS

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Professional Summary

Engineer with 2+ years of experience working on data-intensive systems, with recent experience building GenAI-powered applications including RAG-based LLM solutions and end-to-end ML pipelines. Experienced in designing scalable, low-latency backend architectures and deploying production-ready services using Docker and AWS.

Technical Skills

Languages: Python, SQL

AI and LLM Systems: LangGraph, LangChain, LLM APIs, RAG(Advanced), Hugging Face, Prompt Engineering

Backend: FastAPI, Async Processing, Pydantic

Cloud and MLOps: AWS (EC2, SageMaker, Lambda, S3), Docker, CI/CD, Git/GitHub

Database and Storage: MySQL, MongoDB, Qdrant, FAISS, Redis

Experience

National Skill Development Corporation

April 2024 – Present

Data Analyst

New Delhi

- Built and deployed a RAG-based support assistant for IT support workflows (LLMs + vector search); improved documentation retrieval and reduced manual lookup by 35–45%
- Reduced ticket resolution time by 14–16% by enabling real-time, context-aware retrieval using hybrid search and re-ranking for precise query-to-context matching
- Boosted resolution efficiency by 4.5% YoY by engineering SQL-based performance models to analyze 60K+ historical CRM records and identify bottleneck trends.
- Improved SLA compliance to 93% by performing exploratory data analysis (EDA) on high-volume JIRA datasets and implementing statistical prioritization models to resolve friction points

VIVO Mobiles India Limited

Aug 2021 – Jan 2022

Research Engineer

Delhi NCR

- Increased operational output by 4% through the design and standardization of quantitative KPI metrics derived from historical performance trends.
- Accelerated executive decision-making for 20+ senior stakeholders by developing dynamic Excel dashboards utilizing advanced lookups, pivot tables, and slicers.

Projects

Regulatory Compliance Assistant (RAG) | *LangChain, LLM, Vector Database, FastAPI, AWS*

[Live Demo](#)

- An LLM-based compliance assistant using Qdrant DB, BM25 search and Hybrid re-ranking: achieved <2s latency with clause-level grounding of over 2.5K+ data chunks
- Optimized retrieval quality via metadata filtering and re-ranking; improving top-k relevance by 20–25%, minimizing hallucinations
- Architected and deployed a production-grade GenAI stack (FastAPI, Vite, Docker, AWS EC2, CI/CD) → supported scalable, real-time inference with end-to-end ownership

Credit Default Risk Prediction Model | *Python, XGBoost, SHAP*

[Live Demo](#)

- Achieved 0.78 AUC-ROC and 76% accuracy in predicting customer default probabilities by architecting a supervised learning pipeline using XGBoost on structured financial data.
- Improved model robustness and captured granular spending volatility by engineering 15+ complex features, including Credit Utilization Ratios and Payment-to-Balance trends.
- Enhanced model robustness by integrating Explainable AI (SHAP) to isolate key risk drivers like payment delays and credit utilization spikes, providing transparency into "Black Box" predictions.

Education

Bachelor of Technology in Electronics and Communication Engineering

July 2018 – July 2022

Dr. A.P.J. Abdul Kalam Technical University

Lucknow, U.P.

Certification in Data Science and Machine Learning

Sep 2022 – March 2024

Scaler

Bengaluru